Anmol Agarwal

Noida, India | anmolagarwal2625@gmail.com
linkedin.com/in/anmolagarwal26 | github.com/iAnmolAgarwal | Codeforces: i_anmolagarwal | CodeChef: i_anmolagarwal

Education

Jaypee Institute of Information Technology (JIIT) — Noida, India
B.Tech in Computer Science and Engineering, CGPA: 8.5/10 (Semester 4 SGPA: 9.4/10) — *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Theory of Computation. **In Progress:** Operating Systems, Computer Networks.
- **Class XII:** 92.8% (3rd in school, PCM stream).

Achievements

- **HCLTech AMPLIfied (Season 1):** National hackathon winner — **#1 overall** among **4,500+** participants from 97 colleges; scored 100.00 on the Round 1 ML leaderboard.
- **HackerRank Orchestrate (24-hour global hackathon): #6 of 3,062** worldwide (top 0.2%) in September, after top 4% finishes in June (#63 of 1,773) and August (#63 of 1,983).
- **KIET CODE RUSH 3.0: 1st place** at an offline inter-college programming contest (530+ registrations) — won a 32-team knockout (32 to 16 to 8 to 3) and the top-3 final; prize of INR 15,000.
- **Competitive Programming:** Codeforces **Specialist**, 275-day streak; **3,000+** problems solved across Codeforces and CodeChef.
- **IICPC: Top 16%** of **13,800** registered at an offline international coding contest featuring candidates from IITs and NITs.
- **Research:** Co-authored a manuscript on a low-cost IoT gas-sensor array for food-spoilage monitoring, under review at *Sensors and Actuators Reports* (Elsevier); wrote the C++ sensor firmware.
- **Leadership & Certifications:** Student Placement Coordinator, JIIT (Aug 2026 – Present); Anthropic Claude Code 101.

Projects

pathwise | *Next.js, React, TypeScript, PostgreSQL, Python, Anthropic Claude API* | **GitHub** | **Live** — Aug 2026

- AI-powered personalized learning path recommender, ranked **#1** of the 25 prototypes shortlisted to pitch at HCLTech AMPLIfied (a national hackathon): a planner that turns a vague learning goal into an ordered, week-by-week course plan sized to the hours a learner actually has.
- Kept the model out of the decision — it runs the conversation, while plain code picks and orders the courses, so every step can show why it is there; the prerequisite map behind it was cross-checked against the question order of **2.1M** Stack Overflow users.
- Wrote the spec before any code and held every session to it: a generator-evaluator loop for the interface, independent audit agents reviewing work they had not written, and every screen verified in a browser — finished product in two weeks.

Buy or Wait? — AI Affordability Agent | *Python, Anthropic Claude API* | **GitHub** — Sep 2026

- Built an agent that answers "can I afford this?" from a user's bank history, receipts, and chat messages — the model only reads the documents; every amount, date, and payment plan is computed in plain Python, outside the model (**#6 of 3,062** globally at HackerRank Orchestrate, a 24-hour hackathon).
- Finished the build in **2h22m** by writing the contract first, then directing eight Claude subagents against it, one of them an adversarial verifier kept away from the implementation — all 250 requests answered for $0.0139 each, with zero fallbacks or refused rows.

Jbites | *TypeScript, React Native (Expo), Supabase/PostgreSQL, Vercel; in development* | **Demo** — Jun 2026 – Present

- Built a food-ordering app for the campus cafe so students order from their phones instead of queuing at the counter; two concurrent buyers of the last item get exactly one order — a PostgreSQL function locks those rows (`FOR UPDATE`) first.
- Split a mixed order the moment cooking starts so the drink is handed over while the dish is still on the stove; one PostgreSQL function tags the parts and moves the order on together, or neither happens.

AI-Assisted Development

- Four months of building and shipping in Claude Code since May 2026, with automation rules and hooks written to run it.
- Direct the build from problem statement to architecture, then split it across parallel subagents under a lead model that integrates and reviews their work, and gate the result on independent review loops.

Experience

Bharat Space Education Research Centre (BSERC) — Remote
Summer Intern | SkyWatch capstone: **GitHub** — *Jun 2026 – Jul 2026*

- Delivered **SkyWatch**, the capstone of a 6-week applied space-technology internship — a Python pipeline that reads the live public ADS-B aircraft feed and flags unusual flight behavior on a browser map.
- Hardened the alert path: state is persisted atomically (temp file, `fsync`, replace) so a crash mid-write leaves the last good state intact, and every alert passes one triage gate for duplicates, cooldown, and rate limits.

Technical Skills

Languages: C++ (competitive programming, embedded firmware), Python, TypeScript, JavaScript, SQL, HTML, CSS
AI Engineering: Anthropic Claude API & SDK, Claude Code, OpenAI Codex, LLM agents, prompt-injection defense
Web & Backend: Next.js, React, React Native (Expo), REST APIs, PostgreSQL (relational), Supabase
Tools & Practices: Git & GitHub, GitHub Actions (CI), pytest, Playwright, Vercel